

Economic and Policy Drivers of International Salvage Vehicle Trade: Evidence from U.S. BEV and ICEV Exports, 2023–2025

What economic, policy, and bilateral factors shape international U.S. salvage vehicle exports, and how do these relationships differ between BEVs and ICEVs?

Jenny (Jiaqi) Zhao, Dr. Francisco Pares Olguin

University of California, Davis

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Summer Undergraduate Research Fellowship Program

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Introduction

Salvage vehicles are vehicles that have been declared a total loss or otherwise assigned a salvage title, often following collision, flood, or other damage, and subsequently sold through auction markets. In the United States, platforms such as Copart and Insurance Auto Auctions (IAA) connect these vehicles with domestic and international buyers. As domestic electric vehicle (EV) adoption accelerates, the market is showing an increasing share of salvaged vehicles that are electric. Salvaged EVs also carry high-value lithium-ion battery packs and critical minerals that may retain economic value even after collision or flood damage. Depending on the destination country's policy environment and market incentives, a salvaged EV may enter as a repairable consumer vehicle or as a source of recoverable battery materials.

International trade in used vehicles is well established, with longstanding trade networks connecting major vehicle-exporting countries (United States) with secondary markets abroad. These trade flows have historically been dominated by used and salvaged internal combustion engine vehicles (ICEVs). However, the growing availability of used and salvaged EVs introduces a potentially different set of trade dynamics. Unlike ICEVs, EVs depend on battery technology, charging infrastructure, repair capacity, and supportive policy environments, while their batteries may retain substantial value independent of the vehicle itself. As EVs increasingly enter established used-vehicle trade channels, an important question is whether the same factors that support traditional ICEV trade also shape the emerging international market for salvaged EVs.

This study examines the economic, policy, and bilateral factors associated with international salvage vehicle imports using a country-year panel dataset covering 48 destination countries from 2023 to 2025. Separate panel regressions are estimated for battery electric vehicle (BEV) and internal combustion engine vehicle (ICEV) imports to investigate whether the two markets are associated with different destination-country characteristics.

The key explanatory variables include the IEA Overall EV Policy Index, GDP per capita, U.S.-origin remittances, U.S. foreign aid as a percentage of GDP, and total bilateral trade with the United States. These variables capture differences in national EV policy environments, economic development, financial and institutional linkages with the United States, and broader bilateral trade integration. Comparing the BEV and ICEV models provides a benchmark for assessing whether the emerging salvage EV market follows the same international trade patterns as the more established ICEV market.

Five hypotheses guide the empirical analysis. First, countries with stronger EV policy environments are expected to import more salvage BEVs. Second, higher GDP per capita is expected to be positively associated with salvage BEV imports. Third, stronger financial linkages with the United States, measured through remittance flows, are expected to increase vehicle imports. Fourth, stronger institutional ties with the United States (foreign aid) are expected to be positively associated with imports of both BEVs and ICEVs. Fifth, greater bilateral trade with the United States is expected to be positively associated with salvage vehicle imports, reflecting stronger established trade relationships between the United States and destination markets. The analysis then compares these relationships across BEVs and ICEVs to determine whether the emerging salvage EV market follows the same drivers as the conventional salvage vehicle market.

Existing Literature Review

2.1 Global Used Vehicle Trade (UNEP)

UNEP (2020) finds substantial variation in used-vehicle import regulations across countries, with 81 of 146 countries classified as having “weak” or “very weak” regulatory frameworks and only 47 classified as having “good” or “very good” policies. Regulatory approaches include vehicle age restrictions, emissions standards, emissions- or engine-based taxation, labeling requirements, and selective import bans. Importantly, some countries have also introduced tax exemptions or other incentives for hybrid and electric vehicles, while others that previously prohibited used-vehicle imports have begun allowing used hybrid or fully electric vehicles. UNEP argues that stricter import regulations are generally associated with cleaner and more efficient vehicle technologies entering destination markets. These findings suggest that national policy environments may influence not only the volume of used vehicle imports but also the types of vehicles entering secondary markets (UNEP, 2020). However, UNEP’s analysis primarily reflects the global used-vehicle market before the rapid expansion of EV adoption in the 2020s. It therefore provides limited evidence on whether national policy environments specifically influence the emerging international market for used BEVs or whether the factors associated with BEV imports differ from those associated with conventional ICEV imports.

2.2 Emerging Market for Used Electric Vehicles

The International Transport Forum (ITF) emphasizes that second-hand electric vehicle trade may play an important role in the global transition toward transportation electrification. Emerging economies are expected to depend partly on imported used EVs to decarbonize their vehicle fleets, making international secondary markets an important pathway for expanding access to electric mobility. The report also notes that international used-vehicle trade remains underreported, creating challenges for tracking vehicle flows and enforcing environmental regulations. In the United States, the salvage market represents a major source of vehicles entering international used-vehicle export flows, making salvage auctions an important channel through which U.S. vehicles reach overseas markets. However, existing research primarily examines the broader second-hand vehicle market rather than the salvage vehicle market specifically. Salvaged EVs represent a distinct segment because vehicles sold through salvage auctions may be repaired and returned to service, dismantled for components, or retain additional value through their battery systems. This gap motivates a closer examination of U.S. salvage

vehicle exports and whether the characteristics associated with their international destinations differ between BEVs and ICEVs.

2.3 Battery Circularity and Salvaged EVs

The distinction between electric and conventional vehicles further motivates separate analysis. Unlike internal combustion engine vehicles (ICEVs), the value of a salvaged EV is determined not only by the vehicle itself but also by the remaining value of its battery. Martinez-Laserna et al. (2018) demonstrate that retired EV batteries continue to have economic value through second-life applications and material recovery, suggesting that the international market for salvaged EVs may respond to different economic incentives than the market for conventional vehicles. As a result, factors influencing the international trade of used BEVs may differ from those affecting used ICEVs.

Despite growing interest in international used vehicle markets, important research gaps remain. Much of the existing literature predates the rapid growth of global EV adoption after 2020, and relatively few empirical studies compare the determinants of used battery electric vehicle (BEV) and internal combustion engine vehicle (ICEV) imports across countries. Moreover, limited research jointly examines how national EV policy environments and broader economic factors—including income, remittances, and foreign aid—influence international used vehicle trade. This study addresses these gaps by using a cross-country panel dataset from 2023–2025 to examine the economic and policy drivers of BEV and ICEV imports and to determine whether these two markets respond differently to national characteristics.

2.4 Economic and Bilateral Drivers of International Trade

The gravity model of international trade (Tinbergen, 1962; Anderson, 1979; Anderson & van Wincoop, 2003) provides a useful framework for understanding cross-border used-vehicle flows. In its basic form

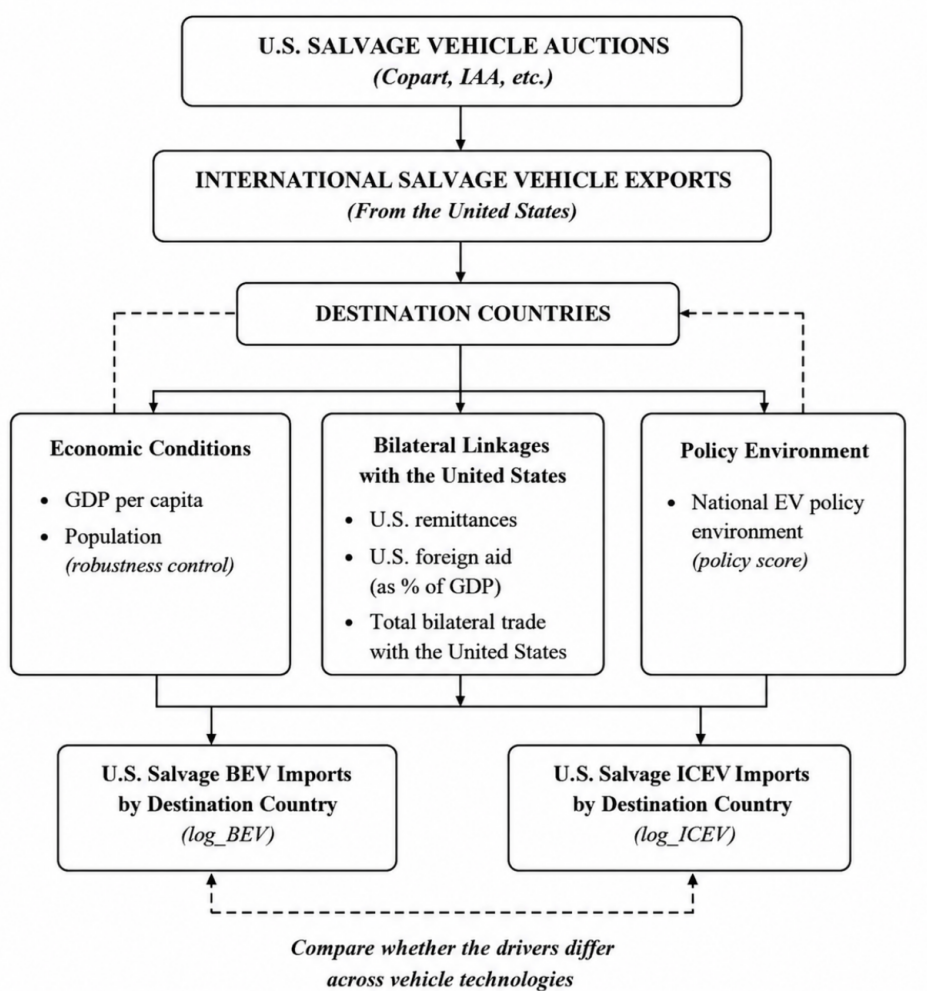
$$T_{ij} = \frac{GDP_i^\alpha GDP_j^\beta}{D_{ij}^\theta},$$

where bilateral trade T_{ij} increases with the economic size of the exporting and importing countries and decreases with trade costs represented by distance D_{ij} . Because this study focuses exclusively on salvage vehicles exported from the United States, variation on the exporter side is largely fixed. The empirical analysis therefore focuses on destination-country economic and policy conditions, as well as broader social, institutional, and commercial linkages with the United States. Remittances serve

as a proxy for migrant-network linkages with the United States, as migrant communities may facilitate trade through information, cultural, and commercial connections. U.S. foreign assistance captures a different dimension of bilateral relationships, reflecting non-commercial institutional ties such as development partnerships, strategic alignment, and broader U.S. engagement with destination countries. Lastly, total bilateral trade captures the strength of established commercial ties between the United States and each destination country.

Methods

3.1 Conceptual Framework



This study examines how destination-country characteristics and linkages with the United States are associated with international imports of salvaged battery electric vehicles (BEVs) and internal combustion engine vehicles (ICEVs). The conceptual framework considers three broad dimensions: economic conditions, bilateral linkages with the United States, and the national EV policy environment. Economic conditions capture differences in market capacity across destination countries, while bilateral linkages capture social, institutional, and commercial connections with the United States through remittances, foreign assistance, and total bilateral trade. The International Energy Agency (IEA) Overall EV Policy Index captures differences in national policy environments supporting EV adoption.

The analysis then compares whether these factors are associated differently with salvage BEV and ICEV imports. EVs differ from conventional ICEVs because they rely on battery technology, charging infrastructure, specialized repair capacity, and EV-supportive market conditions. These differences may cause destination-country characteristics to matter differently for the two vehicle types. Estimating separate models for BEV and ICEV imports therefore allows the study to assess whether the emerging international salvage EV market is driven by the same factors as the more established ICEV market.

3.2 Data Collection

Import Flow Data

Salvaged vehicle import flow data are obtained from AutoAstat, an industry database tracking international vehicle trade originating from US salvage auctions including Copart and Insurance Auto Auctions (IAA). The dataset provides annual import volumes (unit quantity) and total auction sale values in current US dollars for both electric vehicles and internal combustion engine vehicles separately, by destination country, for 2023, 2024, and 2025. This study analyzes BEV and ICEV imports separately to examine whether the two markets respond differently to national economic and policy conditions.

IEA EV Policy Index

Data on national EV policies are drawn from the International Energy Agency (IEA) Global EV Policy Explorer, which provides qualitative information on EV-related policies across countries. Using these policy data, we construct a country-level EV policy score to provide a standardized measure of the strength of each destination country's EV policy environment.

The score is based on a weighted count of active EV-supportive policies. Greater weight is assigned to legally binding and national-level measures, while the scoring framework distinguishes among policy mechanisms such as EV-related taxation, charging infrastructure, reductions in import costs, and EV adoption targets. Higher scores therefore indicate a more extensive and institutionally established policy environment supporting EV adoption.

This constructed measure allows EV policy conditions to be compared consistently across destination countries and incorporated into the panel regression analysis.

Country-Level Control Variables

To account for differences across destination countries, the analysis incorporates measures of economic capacity and bilateral linkages with the United States. GDP per capita (World Bank) captures differences in income and purchasing power. U.S.-origin remittances (KNOMAD/World Bank) proxy for the size and strength of migrant-linked economic networks, while U.S. foreign assistance as a percentage of recipient GDP captures broader institutional and strategic ties with the United States. Total bilateral trade (UN ComTrade) measures overall commercial integration between each destination country and the United States. These variables capture destination-country economic conditions and different dimensions of cross-border connectivity that may shape international salvage vehicle flows.

Table 1. Variable Definitions, Measurement, and Data Sources

Variable	Description	Measurement / Construction	Source	Expected Relationship with BEV Imports
$\log_BEV_{it}$	Salvage BEV import quantity	Natural log of annual salvaged BEV import quantity: $\ln(BEV_{it} + 1)$	<i>AutoAstat</i> (2026)	Dependent variable
$\log_ICEV_{it}$	Salvage ICEV import quantity	Natural log of annual salvaged ICEV import quantity: $\ln(ICEV_{it} + 1)$	<i>AutoAstat</i> (2026)	Dependent variable (comparison)
$EV_policy_score_{it}$	EV policy environment	Constructed score of active EV-supportive policies (higher score = stronger policy environment) ^a	<i>IEA Global EV Policy Explorer</i> (2026)	+
$\log_GDPpc_{it}$	Economic capacity	Natural log of GDP per capita (current USD)	<i>World Bank, World Development Indicators</i> (NY.GDP.PCAP.CD) (2026)	+
$\log_remittances_{it}$	Financial linkages through migrant networks	Natural log of estimated U.S.-origin remittances received (current USD) ^b	<i>World Bank/KNOMAD Bilateral Remittance Matrix</i> (2026)	+
$US_foreign_aid_pct_GDP_{it}$	Institutional and strategic linkages	U.S. foreign assistance received as a percentage of recipient-country GDP	<i>ForeignAssistance.gov</i> (2026) + <i>World Bank WDI</i> (GDP)	+
$\log_bilateral_trade_{it}$	Commercial linkages with the United States	Natural log of total bilateral merchandise trade between the United States and the destination country (U.S. exports to destination + destination exports to U.S., current USD)	<i>UN Comtrade Database</i> (2026)	+ (stronger for ICEV expected)
$\log_population_{it}$	Market size (robustness control)	Natural log of total population (mid-year)	<i>World Bank, World Development Indicators</i> (SP.POP.TOTL) (2026)	+

^a The EV policy score is a researcher-constructed weighted index based on the IEA Global EV Policy Explorer. Policies include areas such as vehicle incentives, taxation, charging infrastructure, fuel economy standards, and EV manufacturing support. Legally binding national policies receive greater weight than non-binding announcements.

^b Bilateral remittances are specifically from the United States to the destination country.

3.3 Country Selection

The empirical analysis uses a balanced panel dataset covering 2023–2025. Countries are included whenever data are available across the required economic and policy variables. Countries with missing information for key variables are excluded only when necessary for model estimation.

3.4 Research Hypotheses

Hypothesis 1 — EV Policy Environment. Countries with stronger national EV policy environments are expected to import greater quantities of salvaged BEVs. This relationship is expected to be weaker for ICEV imports.

Hypothesis 2 — Economic Development. Higher GDP per capita is expected to be positively associated with salvaged BEV imports, reflecting greater purchasing power and capacity to support EV use. The relationship is expected to be weaker for ICEV imports.

Hypothesis 3 — Financial Linkages. Greater U.S.-origin remittance flows are expected to be associated with higher imports of used vehicles, including both BEVs and ICEVs, by capturing financial and migrant-network linkages with the United States.

Hypothesis 4 — Institutional Linkages. Greater U.S. foreign aid relative to recipient-country GDP is expected to be positively associated with salvage vehicle imports. Greater

U.S. foreign assistance relative to recipient-country GDP is expected to be positively associated with salvage vehicle imports. Foreign assistance serves as a proxy for broader institutional ties and political alignment with the United States, which may facilitate stronger bilateral economic relationships and cross-border trade.

Hypothesis 5 — Bilateral Trade Linkages. Greater bilateral trade with the United States is expected to be positively associated with salvage vehicle imports, as stronger existing commercial relationships may facilitate cross-border vehicle flows. This relationship is expected to be stronger for ICEVs than for BEVs if conventional salvage vehicles are more deeply embedded in established U.S. trade networks.

3.5 Empirical Econometric Model

To isolate the determinants of vehicle trade and evaluate market differences, we specify two separate panel regression models for BEVs and ICEVs:

Model (estimated separately for BEVs and ICEVs) $\ln(\text{VehicleImports}_{it} + 1) = \beta_0 + \beta_1 \text{EVPolicy}_{it} + \beta_2 \ln(\text{GDPpc}_{it}) + \beta_3 \ln(\text{Remittances}_{it}) \\ + \beta_4 \text{ForeignAid}_{it} + \beta_5 \ln(\text{BilateralTrade}_{it}) + \gamma_t + \epsilon_{it}$

Where:

- i i indexes 48 destination countries and t indexes years 2023–2025.
- ii VehicleImports_{it} represents annual U.S. salvage BEV or ICEV imports, with separate models estimated for each vehicle type.
- iii EVPolicy measures the national EV policy environment
- iv GDPpc captures destination-country income
- v Remittances proxy migrant and financial linkages with the United States
- vi ForeignAid captures institutional and political ties
- vii BilateralTrade measures overall commercial ties with the United States.
- viii Year fixed effects (γ_t) account for common time-specific shocks, and standard errors are clustered at the country level.

Empirical Results & Descriptive Findings

4.1 Geographic Distribution of Salvage BEV/ICEV Market Readiness

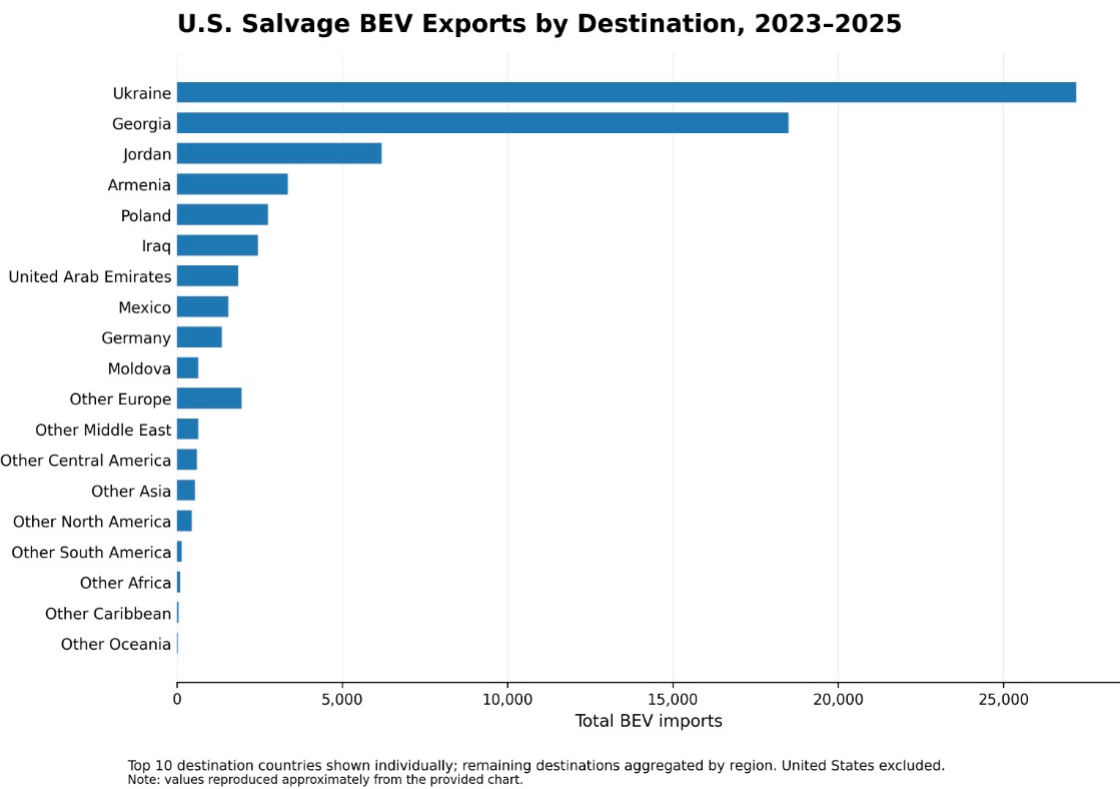


Figure 2. Geographic Distribution of Used BEV Imports (2023–2025).

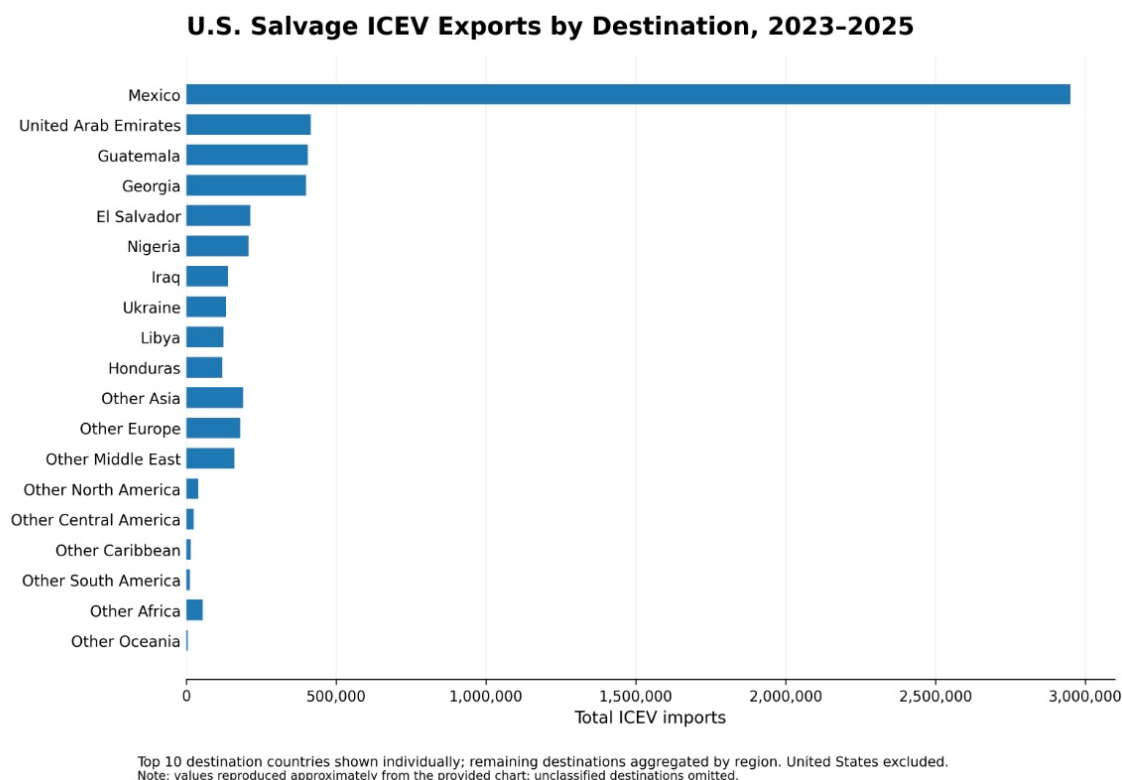


Figure 3. Geographic Distribution of Used ICEV Imports (2023–2025).

Figures 2 and 3 present the distribution of U.S. salvage BEV and ICEV exports across destination markets from 2023 to 2025. The ten largest destination countries are shown individually, while the remaining destinations are aggregated by region to provide a clearer comparison of the geographic concentration of the two salvage vehicle markets.

The figures reveal distinct trade patterns across vehicle technologies. BEV exports are concentrated primarily in Eastern Europe and nearby markets, with Ukraine and Georgia among the largest destinations. In contrast, ICEV exports are heavily concentrated in Mexico, which substantially exceeds other destination countries. Beyond these leading markets, both vehicle types are distributed across several regions, but with noticeably different patterns of concentration. These descriptive differences suggest that BEV and ICEV salvage trade may operate through different international market channels, motivating the separate regression models used in the following analysis.

4.2 Summary Statistics

Table 2. Summary Statistics of Key Variables (\$N = 144\$)

Variable	Mean	Std. Dev.	Min	Max
BEV Imports (Units)	1,428.3	2,891.5	12	10,840
ICEV Imports (Units)	18,452.1	34,102.8	185	142,500
EV Policy Score	24.15	18.62	0.00	78.50
GDP per capita (USD)	12,450.8	14,210.3	680.2	65,400.0
Remittances (USD Millions)	1,120.4	2,450.1	2.1	14,800.0
U.S. Foreign Aid (% of GDP)	0.42	0.88	0.00	4.85
Repair Labor Cost (USD/hr)	7.85	6.72	0.35	22.83

Table 2 presents summary statistics for the main variables used in the analysis. The descriptive statistics show a substantial difference in the scale of the two salvage vehicle markets. Average ICEV imports are 18,452 units per country-year, compared with 1,428 BEV imports, indicating that the conventional salvage vehicle market remains considerably larger. Both import measures also exhibit substantial variation across observations. Destination countries differ considerably in their economic and policy characteristics: EV policy scores range from 0 to 78.5, while GDP per capita ranges from approximately \$680 to \$65,400. This variation provides a useful basis for examining whether economic conditions, bilateral linkages, and EV policy environments are associated differently with BEV and ICEV import volumes.

Table 3. Characteristics of Major Salvage Vehicle Destination Markets, 2023–2025

Country	BEV Imports 2023–2025 (units)	ICEV Imports 2023–2025 (units)	GDP per Capita (USD)	EV Policy Score (average)	Remittances (USD millions)	U.S. Foreign Aid (% of GDP)	Bilateral Trade (USD billions)
Ukraine	27,304	133,015	5,466	1.66	3.0	16.08%	2.17
Georgia	18,548	398,929	8,981	1.42	0.0	0.43%	1.92
Jordan	6,254	34,486	5,109	0.83	3.5	2.91%	3.98
Armenia	3,338	66,940	8,730	2.98	0.0	0.29%	2.14
Poland	2,742	53,530	25,223	8.52	47.1	0.32%	24.83
Mexico	1,520	2,947,849	13,903	3.87	2,214.3	1.04%	83.97
Germany	1,398	34,344	57,126	3.76	305.4	0.02%	24.79
United Arab Emirates	1,088	263,942	50,062	1.80	0.0	0.01%	32.69
Moldova	613	9,649	7,667	0.00	0.0	0.98%	1.34
Canada	503	39,437	55,187	3.15	640.6	0.00%	76.84
El Salvador	274	213,400	5,530	1.65	15.3	0.51%	6.67
Guatemala	156	403,576	6,168	1.00	18.9	0.18%	14.61
Nigeria	107	209,020	1,482	2.11	0.0	0.26%	8.49
Honduras	38	120,975	3,414	0.46	22.5	0.48%	9.11
India	15	62,712	2,576	5.46	115.6	0.38%	12.51

Notes: BEV and ICEV imports are cumulative totals for 2023–2025. All other variables are country-level averages over 2023–2025.

Sources: AutoAstat; World Bank (World Development Indicators); EV Policy Score Dataset; World Bank (Migration and Remittances); OECD (Creditor Reporting System); UN Comtrade.

Table 3 highlights substantial differences between the major destination markets for U.S. salvage BEV and ICEV exports. BEV exports are particularly concentrated in Ukraine and Georgia, while Mexico is by far the largest ICEV destination, receiving nearly 2.95 million vehicles over the 2023–2025 period.

Importantly, the largest BEV destinations are not necessarily countries with the highest GDP per capita or EV policy scores. Ukraine and Georgia have relatively modest income levels and policy scores but receive substantially more salvage BEVs than wealthier markets such as Germany and Canada. The bilateral indicators also vary across major destinations: Mexico combines exceptionally high ICEV imports with very large bilateral trade and remittance linkages with the United States, whereas Ukraine combines high BEV imports with comparatively smaller commercial trade but substantially greater U.S. foreign assistance relative to GDP.

Overall, these descriptive patterns suggest that the international salvage vehicle market cannot be explained by economic development or EV policy alone. Instead, the importance of broader U.S.–destination-country linkages appears to differ across markets and vehicle technologies, motivating the multivariate regression analysis that follows.

4.3 Main Panel Regression Results

Table 4 presents the panel regression results, which show several important differences between the BEV and ICEV markets. U.S. foreign aid is positively and significantly associated with both BEV ($\beta = 0.3564$, $p < 0.001$) and ICEV imports ($\beta = 0.1934$, $p < 0.001$). Because foreign aid is used here as a proxy for broader institutional linkages with the United States, the result is consistent with—but does not establish—the possibility that broader bilateral relationships are associated with salvage vehicle trade.

GDP per capita shows notably different relationships across vehicle types. It is positively associated with BEV imports ($\beta = 0.3595$, $p < 0.10$) but negatively associated with ICEV imports ($\beta = -0.8402$, $p < 0.01$). This suggests that higher-income destination countries may be relatively more conducive to salvage BEV demand, while salvage ICEVs are more concentrated in lower-income markets. Bilateral trade is strongly and positively associated with ICEV imports ($\beta = 0.8045$, $p < 0.001$), but its relationship with BEV imports is smaller and statistically insignificant. Finally, the national EV policy score is not statistically significant in either model, providing little evidence that formal EV-supportive policies alone explain observed salvage BEV imports.

Table 4. Baseline Panel Regression Results for Salvage BEV and ICEV Imports

Variable	BEV Imports (Dependent variable: log units)	ICEV Imports (Dependent variable: log units)
U.S.-origin remittances (log)	-0.1063** (0.0355)	-0.1284* (0.0495)
GDP per capita (log)	0.3595† (0.1816)	-0.8402** (0.2965)
National EV policy score	0.0306 (0.0977)	-0.1961 (0.1437)
U.S. foreign aid (% of GDP)	0.3564*** (0.0469)	0.1934*** (0.0390)
Bilateral trade with the U.S. (log)	0.1966 (0.1296)	0.8045*** (0.2117)
Year fixed effects	Yes	Yes
Country-clustered standard errors	Yes	Yes
Observations	144	144
R^2	0.2915	0.2312
Within R^2	0.2852	0.2275

Notes: Standard errors are in parentheses and clustered at the country level.

*** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, † $p < 0.10$.

4.4 Comparison of BEV and ICEV Drivers

Figure 4. Estimated Drivers of U.S. Salvage BEV and ICEV Imports.

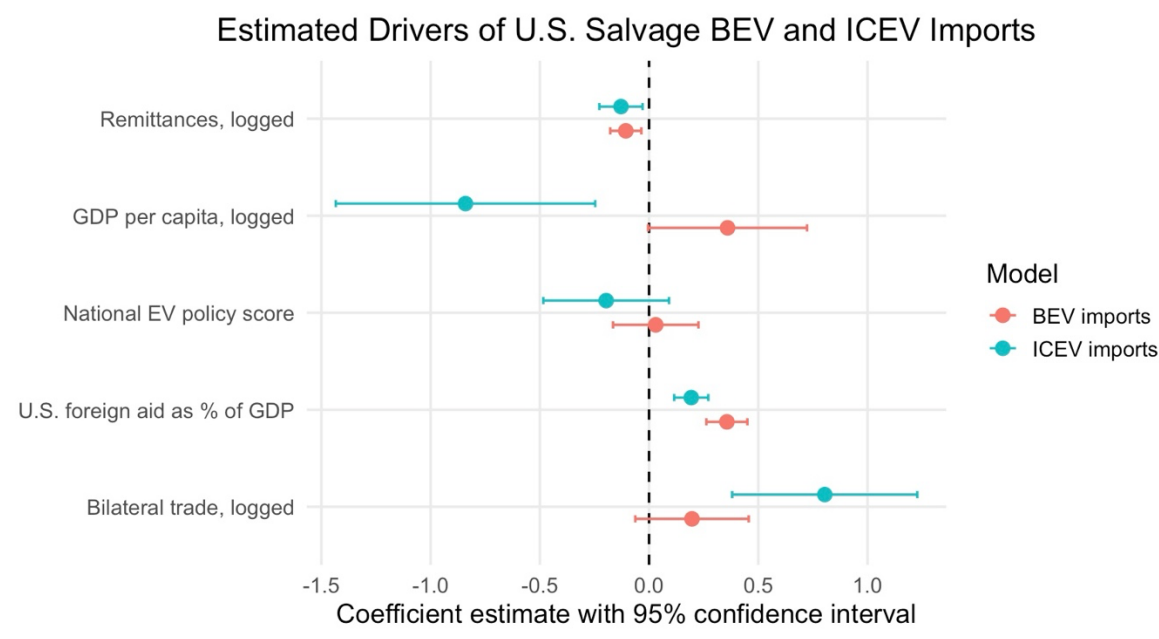


Figure 4 visually reinforces the differences between the two markets. The largest contrast appears in GDP per capita and bilateral trade, while U.S. foreign aid shows a consistent positive relationship across both vehicle types. Overall, the figure supports the conclusion that salvage BEV and ICEV imports are associated with different destination-country characteristics, rather than following a single common trade pattern.

Table 5. Summary of Empirical Findings Across Vehicle Types

Factor	BEV Result	ICEV Result	Interpretation
Economic Development (GDP per capita)	Positive, marginally significant ($\beta = 0.3595, p < 0.10$)	Negative, significant ($\beta = -0.8402, p < 0.01$)	Higher-income countries tend to import more salvage BEVs but fewer salvage ICEVs, indicating different demand patterns across vehicle technologies.
U.S. Institutional Ties (Foreign Aid)	Positive, significant ($\beta = 0.3564, p < 0.001$)	Positive, significant ($\beta = 0.1934, p < 0.001$)	Stronger U.S. institutional and political ties are associated with greater salvage imports of both vehicle types.
Bilateral Trade with the U.S.	Positive, insignificant ($\beta = 0.1966, p = 0.133$)	Positive, significant ($\beta = 0.8045, p < 0.001$)	Established U.S. commercial ties are strongly associated with ICEV salvage imports, while the relationship is weaker for BEVs.
National EV Policy	Positive, insignificant ($\beta = 0.0306, p = 0.754$)	Negative, insignificant ($\beta = -0.1961, p = 0.174$)	EV policy strength does not independently predict salvage imports of either vehicle type in the current models.
U.S.-origin Remittances	Negative, significant ($\beta = -0.1063, p < 0.01$)	Negative, significant ($\beta = -0.1284, p < 0.05$)	Stronger remittance linkages are associated with lower salvage vehicle imports after controlling for other country characteristics.

Notes: Results are from panel regressions with year fixed effects and country-clustered standard errors. Significance levels:

*** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, + $p < 0.10$.

Figure 5. Bilateral Trade and U.S. Salvage ICEV Imports, 2023–2025

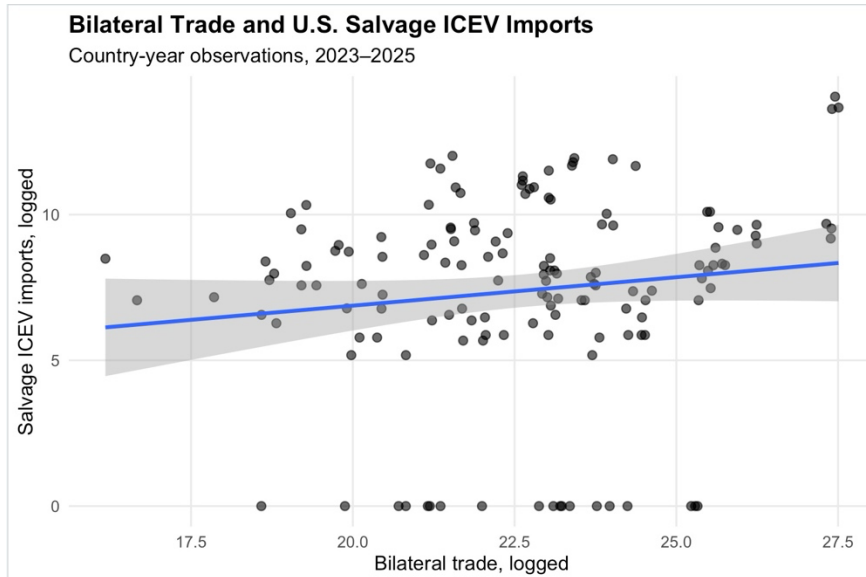
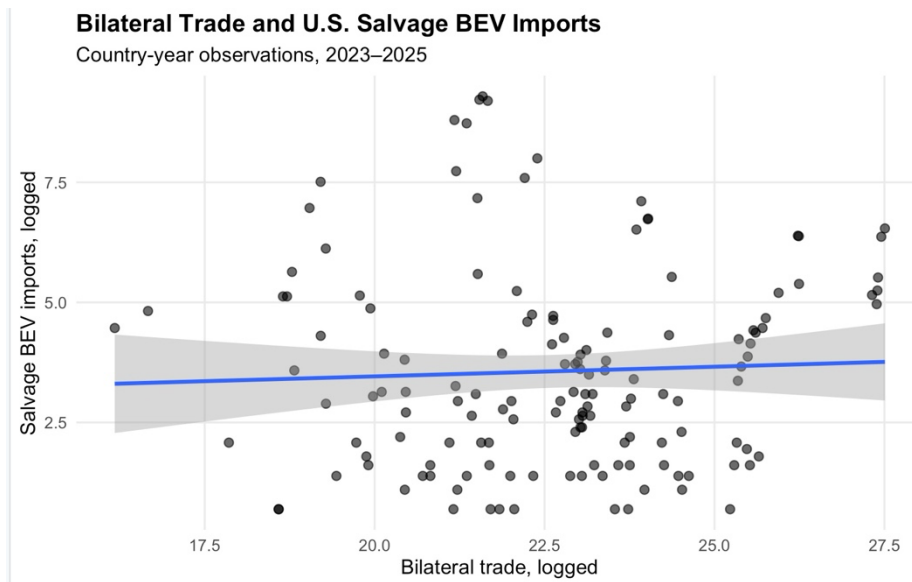


Figure 6. Bilateral Trade and U.S. Salvage BEV Imports, 2023–2025



Figures 5 and 6 provide a descriptive comparison of the relationship between bilateral trade with the United States and salvage vehicle imports.

The ICEV figure shows a clearer positive relationship, with countries that have greater bilateral trade generally importing larger quantities of salvage ICEVs. In contrast, the relationship for BEVs is substantially flatter and more dispersed, suggesting that broader trade integration with the United States is less predictive of salvage BEV flows.

These descriptive patterns are consistent with the multivariate results in Table 4, where bilateral trade is strongly and positively associated with ICEV imports ($\beta = 0.8045$, $p < 0.001$) but is smaller and statistically insignificant for BEV imports ($\beta = 0.1966$). Taken together, the results suggest that conventional salvage vehicle trade may operate through established U.S.–destination-country commercial networks, while the international market for salvage BEVs may depend on additional market-specific factors beyond overall bilateral trade.

4.5 Population Robustness Check

Because the dependent variables measure total import quantities rather than imports per capita, an additional specification controls for destination-country population. Table 5 compares the preferred models with otherwise identical specifications that include logged population. Population itself is not statistically significant for either BEV or ICEV imports, and its inclusion produces only modest improvements in overall model fit.

Several central findings remain robust. U.S. foreign aid remains positively and significantly associated with both BEV and ICEV imports, while remittances remain negatively associated with both. The national EV policy score remains statistically insignificant. Bilateral trade also remains an important predictor of ICEV imports and becomes marginally significant for BEVs after controlling for population. However, the BEV coefficient on GDP per capita becomes substantially smaller and statistically insignificant, indicating that the previously observed positive relationship between income and BEV imports is not robust to controlling for destination-country population. Overall, the results suggest that market size alone does not explain cross-country differences in salvage vehicle imports, although accounting for population affects some estimated economic relationships.

Variable	BEV Main	BEV + Population	ICEV Main	ICEV + Population
U.S.-origin remittances (log)	-0.1063** (0.0355)	-0.1171** (0.0364)	-0.1284* (0.0495)	-0.1433** (0.0475)
GDP per capita (log)	0.3595† (0.1816)	0.0541 (0.3335)	-0.8402** (0.2965)	-1.264* (0.4955)
National EV policy score	0.0306 (0.0977)	0.1000 (0.1248)	-0.1961 (0.1437)	-0.0999 (0.1786)
U.S. foreign aid (% GDP)	0.3564*** (0.0469)	0.3724*** (0.0442)	0.1934*** (0.0390)	0.2156*** (0.0425)
Bilateral trade with U.S. (log)	0.1966 (0.1296)	0.3955† (0.2070)	0.8045*** (0.2117)	1.081** (0.3509)
Population (log)	—	-0.4106 (0.3240)	—	-0.5699 (0.4984)
Year fixed effects	Yes	Yes	Yes	Yes
Country-clustered SEs	Yes	Yes	Yes	Yes
Observations	144	144	144	144
R ²	0.2915	0.3126	0.2312	0.2462
Within R ²	0.2852	0.3065	0.2275	0.2425

Discussions

The regression results reveal an important contrast between formal EV policy and broader structural and bilateral conditions. Contrary to the first hypothesis, the national EV policy score is not significantly associated with salvage BEV imports. Instead, several economic and bilateral variables show stronger relationships with observed trade patterns. GDP per capita is positively associated with BEV imports in the main specification, U.S. foreign aid is positively associated with both BEV and ICEV imports, and bilateral trade is particularly strongly associated with ICEV imports. These findings raise a broader question: why do economic capacity and existing international linkages appear to explain salvage vehicle trade better than formal national EV policy? Three possible explanations emerge from the empirical results and the structure of international secondary vehicle markets.

5.1 Economic capacity may be a prerequisite for salvage BEVs

First, salvage BEV trade may depend more heavily on underlying economic and technological capacity than on formal EV policy alone. Unlike conventional ICEVs, damaged BEVs may require access to charging infrastructure, reliable electricity, battery diagnostics, specialized repair knowledge, and replacement components. GDP per capita may therefore capture dimensions of market readiness that influence whether imported salvage BEVs can be economically repaired and operated.

The population robustness analysis adds nuance to this interpretation. The positive GDP-per-capita relationship observed in the main BEV specification becomes weaker after controlling for population, while population itself is not statistically significant. Therefore, GDP per capita should not be interpreted as a direct measure of BEV readiness. Rather, the overall pattern is consistent with the possibility that salvage BEV markets depend on a combination of economic capacity, infrastructure, repair capabilities, and other destination-country characteristics that are not fully captured by population or formal EV policy.

5.2 Existing trade networks matter differently for BEVs and ICEVs

Second, the results suggest that different forms of U.S.–destination-country linkage should not be treated as equivalent. Bilateral trade with the United States is strongly and positively associated with ICEV salvage imports, but its relationship with BEV imports is

substantially smaller and not statistically significant in the main specification. This difference is especially informative because it suggests that conventional salvage vehicles may move through established commercial channels: countries that already trade extensively with the United States may have stronger shipping connections, importer networks, logistics infrastructure, and experience sourcing U.S. vehicles.

The much weaker bilateral-trade relationship for BEVs suggests that salvage BEV flows may not yet follow these established trade channels to the same extent. Because the international secondary BEV market is newer and technologically distinct, destinations may require additional conditions—including battery-repair capacity, charging availability, parts access, and technical expertise—before existing trade relationships translate into substantial BEV imports. This may help explain why bilateral trade is a powerful predictor of ICEV imports but not of BEV imports.

U.S. foreign aid provides a second measure of bilateral linkage and is positively associated with both vehicle types. This should not be interpreted as evidence that foreign aid directly finances salvage vehicle purchases. Instead, foreign aid may capture dimensions of broader U.S. institutional, political, or economic engagement that overlap with commercial relationships or reduced informational and transaction barriers.

By contrast, U.S.-origin remittances are negatively associated with both BEV and ICEV imports in the preferred specification. This provides no support for H3 and suggests that household-level financial linkages do not operate in the same way as country-level trade and institutional relationships. The negative coefficients may reflect characteristics of destinations receiving substantial remittance flows rather than a causal negative effect of remittances themselves, and the present analysis cannot distinguish between these mechanisms.

5.3 Formal EV policy may not be designed for the salvage market

Third, the insignificant EV policy coefficient may reflect a mismatch between what national EV policies measure and the particular market examined here. Broad EV policy frameworks generally target new-vehicle adoption, charging infrastructure, purchase incentives, taxation, and emissions objectives. Salvage vehicles operate through a distinct secondary-market channel: vehicles are purchased through auctions, exported internationally, and subsequently repaired, dismantled, reused, or potentially used for parts.

A country can therefore maintain an ambitious EV policy environment without necessarily being an attractive destination for U.S. salvage BEVs. Conversely, countries with relatively modest formal EV policies may still develop active salvage BEV markets when repair economics, vehicle prices, technical capacity, and import regulations make these vehicles commercially viable.

This distinction is consistent with the broader used-vehicle literature. UNEP documents substantial cross-country variation in regulations governing used-vehicle imports, including vehicle-age restrictions, emissions standards, fiscal measures, selective bans, and incentives for used hybrid and electric vehicles. ITF similarly emphasizes the growing importance of international second-hand EV trade for emerging economies while identifying limitations in the monitoring of these flows. These observations suggest that secondary-market-specific policies may be more relevant to salvage BEV trade than broad national EV policy measures.

5.4 Overall Assessment of Research Hypotheses

Overall, the empirical results provide mixed support for the proposed hypotheses. H1 is not supported: the national EV policy score does not significantly predict salvage BEV imports. H2 receives partial support, as GDP per capita is positively associated with BEV imports in the main specification, although this relationship is less robust after the inclusion of population. H3 is not supported, as U.S.-origin remittances are negatively rather than positively associated with salvage imports. H4 receives the strongest support, with U.S. foreign aid positively associated with both BEV and ICEV imports. H5 receives partial support. Bilateral trade with the United States is strongly and positively associated with ICEV imports, but its relationship with BEV imports is smaller and statistically insignificant in the main specification. This suggests that established commercial ties may be considerably more important for conventional salvage vehicle flows than for the emerging salvage BEV market.

The addition of bilateral trade further sharpens the central finding. Established U.S. commercial ties strongly predict ICEV salvage imports but do not explain BEV imports nearly as well. Taken together, the results suggest that the international salvage BEV market is not simply a smaller version of the conventional used-vehicle market. Instead, BEV salvage flows may depend on a distinct combination of economic capacity, technological readiness, repair economics, and secondary-market institutions that broad EV policy indicators and traditional trade relationships do not fully capture.

Conclusion

This study examines whether international U.S. salvage BEV imports are associated with the same destination-country characteristics as conventional salvage ICEV imports. Using a panel of 48 destination countries from 2023 to 2025, the analysis compares the role of national EV policy, economic development, remittance linkages, U.S. foreign assistance, and bilateral trade.

The results indicate important differences between the two markets. National EV policy is not significantly associated with salvage BEV imports, while U.S. foreign aid is positively associated with imports of both vehicle types. Bilateral trade is strongly associated with ICEV imports but substantially less predictive of BEV flows, suggesting that the emerging international salvage BEV market may not yet operate through the same established commercial networks as conventional salvage vehicles. GDP per capita is positively associated with BEV imports in the main specification, although this relationship weakens after controlling for population and should therefore be interpreted cautiously.

These findings suggest that understanding international secondary EV markets requires looking beyond policies designed primarily for new-EV adoption. Used- and salvage-vehicle regulations, battery repair capacity, charging infrastructure, recycling systems, and other secondary-market institutions may be important dimensions for future research. Because the analysis covers only three years, the estimated relationships should not be interpreted causally. Future research could extend the time period, incorporate vehicle-level auction characteristics and shipping costs, and develop measures specifically capturing used-EV import regulation, battery circularity, and repair-market capacity.

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Appendix

